

Scenario & Task

A Small Business, EcoGoods, has recently acquired compute resources on-premises which they wish to begin deployment of their website, product database, and any related services. Your role as systems administrator for this company is to configure an Arch Linux system to meet the requirements of the company laid out below in Requirements. This machine should not have a GUI installed on it at any point; any and all configuration tasks must be completed from a CLI.

For the purposes of this assessment, you should create a single Virtual Machine with a 40GB vDisk split into multiple files, for the purposes of installing Arch Linux. This will act as your baremetal server. For the specifications of this machine, ensure that you configure the VM to have at least 2GB of memory, otherwise installation may fail. For CPU, this can be left at default values.

Requirements

1. Install the Arch Linux OS. In total you should have a 40GB Virtual Disk, as multiple files. Your system should have the following partition configuration:

1. /boot - 512M
2. /home - 14G
3. / - 25.5G (remaining)

2. Secure, with justification, the root user account.

3. Set up administrative users for yourself and the IT Manager (Ashley).

4. Create user accounts where needed for the following persons:

1. Avery Wolfgang Stephenson - Technical Lead Developer, should have complete access to any and all folders under /srv/.
2. Evelyn Hughes - Website developer who will need access to a folder (/srv/http) for the website files only. She will put files in there such as index.php, and any other website-related files/folders. She will not require interacting via CLI.
3. James Hunter - Software Tester responsible for verifying that developed work conforms to company standards and works appropriately.

5. Set up and correctly configure an SSH server for secure access, taking into account all user account requirements.

6. Store, and secure access to, a development project data directory (under /srv/) for developers to have access to. Developers should have full access to the development folders; however, the Software Testers should not be able to change code, only check the code for code-style and whether the program/scripts runs appropriately.

7. Installation, configuration, and running of a Docker containerisation environment. Only Technical Leads and System Administrators should have access to control Docker containers' lifecycles. Note, Avery requires a database admin user.

1. MySQL - Not externally accessible. Only by other containers.
2. PhpMyAdmin - Accessible via reverse proxy on /phpmyadmin route.
3. NGinx - Accessible on Port 80. The website will be hosted on the main "/" route.